

Help for Authors of IEEE TPEL to Improve Papers

Often, authors ask for suggestions on how to improve their chances of having a paper accepted to the *IEEE Transactions on Power Electronics* (TPEL). This is difficult to summarize in a set of rules because the decision depends on the expert reviewers' opinion of the research quality and presentation in the paper. However, we have noticed that the following factors seem to favor a paper:

- 1) Research contribution is clearly explained: A major criteria for the acceptance of a paper is that the research result be deemed original and important and not an incremental contribution to the field. Authors should therefore explicitly **emphasize what is new** in the paper and **how it improves state-of-the-art**. Papers are rated by reviewers on a scale of 1(low) – 10(highest) about their research contribution quality. *Accepted papers often have a rating of 7 or more.*

The most common reason for a paper not to be accepted is that the research contribution was deemed insufficient over the known results in the literature. Sometimes this is due to poor literature survey with poor explanation of how the results relate to prior art. If this is the case, then authors should consider better emphasizing and comparing the novelty of results over the latest known cited research. This is achieved by extensively citing the relevant literature and explaining how they differ from the results achieved in the paper. This should be written both in the introduction of the paper and also emphasized when presenting experimental results. Other times, low research contribution score is caused by insufficient research results, and therefore additional, stronger research results must be added before publication. In either case, most papers cannot be published unless they receive high enough research contribution score (review papers are an exception to this criteria).

- 2) Comparison of results to relevant publications in the literature: The author should be aware that Reviewers are asked to rate the quality of references in a paper (on a scale of 1 to 5), among other categories. It is the authors' responsibility to clearly explain how their results *relate to*
 - a. Recent relevant papers¹ published in the top quality power electronics publications. This allows the reviewer to understand the results in the context of other papers published within the past few years. The power electronics technology field rapidly changes, with efficiencies improving, new topologies invented, and technological advancements in all aspects at a rapid pace. It is vital that the authors compare results with the latest and most advanced research to show their superiority of methods.

TPEL does not advocate unwarranted references or self-citation of TPEL articles. In fact, *the editorial board pledges to not weight TPEL references as being more important than other fully reviewed references.* We are, however, insisting that the research results be compared to and shown to be improvements over relevant state-of-the-art research on similar topics. Also, authors are responsible for comparing results to early access papers that have been posted on a journal's website and are available to authors at a time that was prior to the TPEL submitted manuscript by the author. For example, IEEE TPEL early access papers can be found on our Early Access tab for our Transactions on IEEE Xplore:

<http://ieeexplore.ieee.org/xpl/tocresult.jsp?isnumber=4359240>.

¹ It is preferable to cite the journal version of any paper. Some papers first appear in a conference and then later are improved and are published to a journal. The journal version will normally be an expanded version of the conference paper and would have undergone more rigorous review. Therefore, it is better to cite this version of the paper, if such a journal version of the paper exists.

- b. Historical papers published in the field: It is important to explain, perhaps in the introduction, how the results relate to relevant papers that have strongly influenced the research topic over the past 10 or 15 years (or even more). Recently, there has been so much emphasis in the publication community on citing recent publications that authors forget to cite the most important papers on the topic that may have appeared many years ago¹.

It is a simple observation that reviewers seem to enjoy papers that explain how other work in the field relates to submitted manuscript's research results. Also, it is encouraged that the authors explain drawbacks of their results. This openness is appreciated by the reviewers and the readers.

- 3) Original results: Innovation and originality is emphasized on the review form. Often authors split up their results into multiple papers, and the research contribution becomes more incremental. These papers tend to get reviewed worse. Further, we now electronically verify that there is low similarity between every manuscript submitted and previous published material. If a high similarity index is achieved because of prior publication or dual submission, the manuscript must be rejected. Simply stated, the authors are urged to add more new research results to their papers to increase the chances of publication.
- 4) High quality English: It may even be suitable to have professional proofreading services edit the paper before submission. Authors should understand that the reviewers' have finite amount of time to read a paper, and they will not correct substantial English mistakes. TPEL understands that English may be a second language for many authors. After a paper is accepted, we subscribe to a professional editing service that will mildly improve the papers English and grammar. Unfortunately, the reviewers rate a paper before receiving this English editing service.

Finally, it should be clearly understood that the **editorial board does not "average" reviewer scores or comments.** It is possible that some reviewers rate a paper worth publishing and, yet, only a single reviewer notices technical mistakes. In this case, we would never publish a paper with technical shortcomings. It is the job of the editorial board to thoroughly examine all reviewer comments and then gauge them appropriately.